

Erdős Problem #945

EP-945

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $F(x)$ be the maximal k such that there exist $n+1, \dots, n+k \leq x$ with $\tau(n+1), \dots, \tau(n+k)$ all distinct (where $\tau(m)$ counts the divisors of m). Estimate $F(x)$. In particular, is it true that $F(x) \leq (\log x)^{O(1)}$? In other words, is there a constant $C > 0$ such that, for all large x , every interval $[x, x + (\log x)^C]$ contains two integers with the same number of divisors?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>